UNIVERSITY OF KARACHI

Department of Computer Science (UBIT)

LAB NO. 03

Operating Systems (SE-458)

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LAB NO. 03 - FILE AND DIRECTORY OPERATIONS

PART A: COMMAND REFERENCE

This section provides a summary of basic Linux commands used for file and directory management, text viewing, and system navigation.

Basic Navigation and Directory Commands

- pwd - Print Working Directory: Displays the full path of the current working directory.
- ls - List Directory Contents: Lists files and folders in the current directory. Common options include -l (long listing) and -a (show hidden files).
- cd - Change Directory: Navigates between directories (e.g., cd .. moves up one level, cd ~ moves to the home directory).
- mkdir - Make Directory: Creates new directories (e.g., mkdir folder_name or mkdir -p path/to/dir for nested directories).
- rmdir - Remove Directory: Removes empty directories.

File Manipulation Commands

- touch - Create Empty File: Creates an empty file or updates the timestamp of an existing file.
- cp - Copy Files/Directories: Copies files or directories from source to destination (e.g., cp file1 file2 or cp -r dir1 dir2).
- mv - Move or Rename: Moves files/directories or renames them if the destination is in the same directory.
- rm - Remove Files/Directories: Deletes files or directories (e.g., rm file_name or rm -rf dir_name).

File Viewing and Editing Commands

- cat - Concatenate and Display: Displays the full contents of a file on the terminal.
- head / tail - View Beginning/End: Displays the first or last few lines of a file (default is 10 lines).
- nano / vim - Text Editors: Terminal-based text editors for creating and modifying files.

PART B: CLASSWORK - FILE AND DIRECTORY OPERATIONS

Task 1: File and Directory Navigation

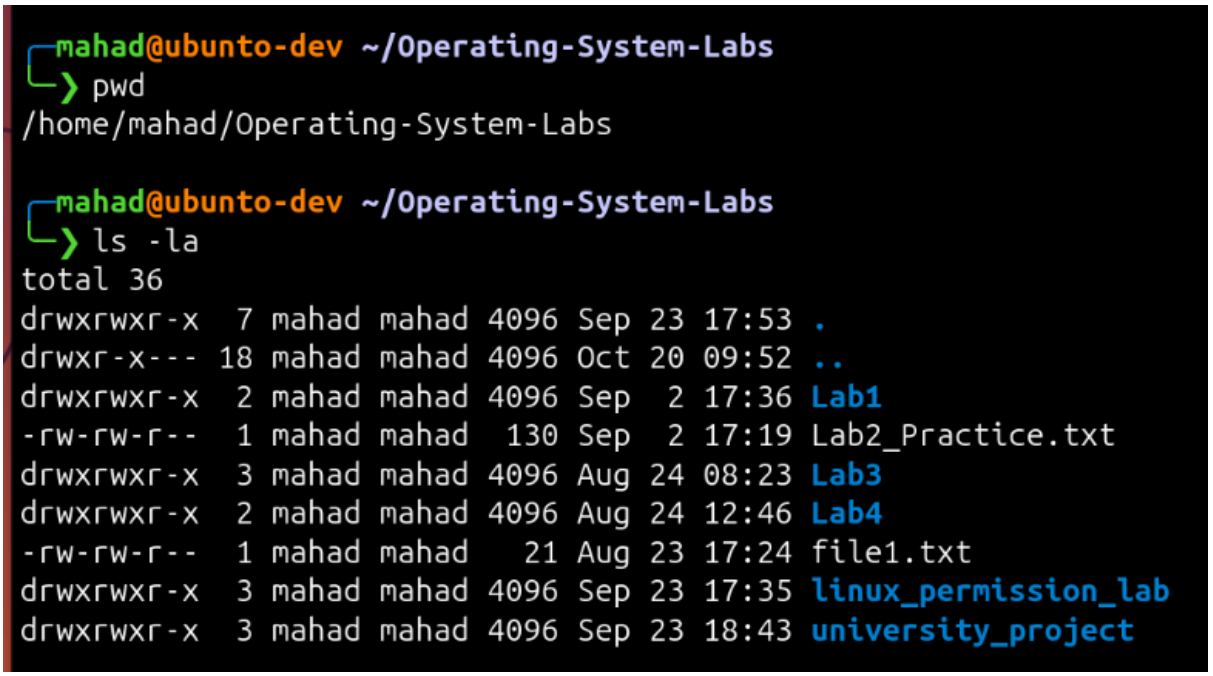
Display the current working directory path, list all files including hidden ones in detailed format, and navigate to the home directory.

Commands:

```shell
pwd
ls -la
cd ~
```

The pwd command prints the absolute path of the current directory. The ls -la command lists all entries including hidden files with detailed metadata. The cd ~ command changes the directory to the user's home folder.

Output:



Task 2: Creating Directory Structures

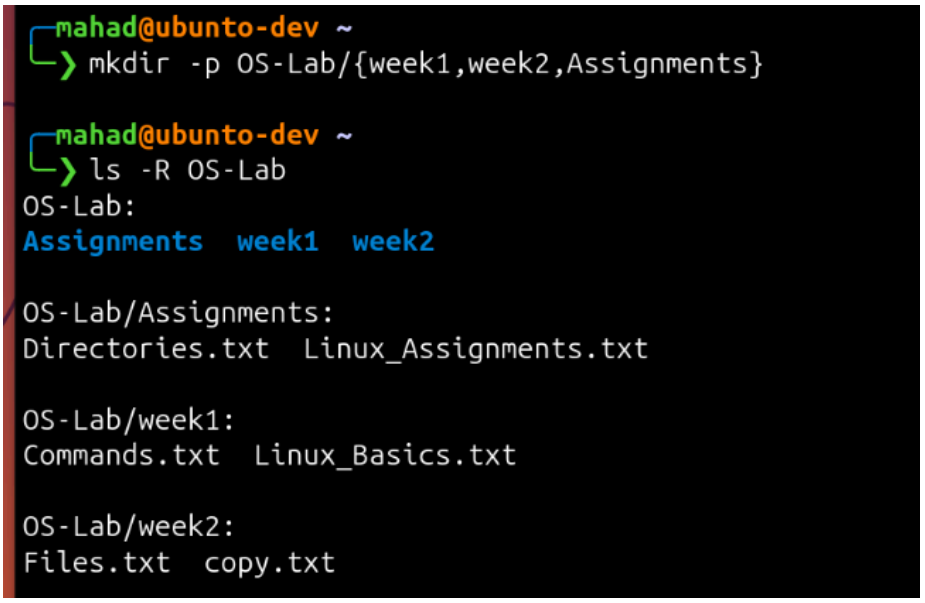
Create a main directory named OS-Lab containing subdirectories week1, week2, and Assignments using a single command, then verify the structure.

Commands:

```shell
mkdir -p OS-Lab/{week1,week2,Assignments}
ls -R OS-Lab
```

The mkdir -p command with brace expansion creates the entire parent and sub-directory hierarchy at once. The recursive ls -R command displays the created tree structure.

Output:



Task 3: File Creation and Management

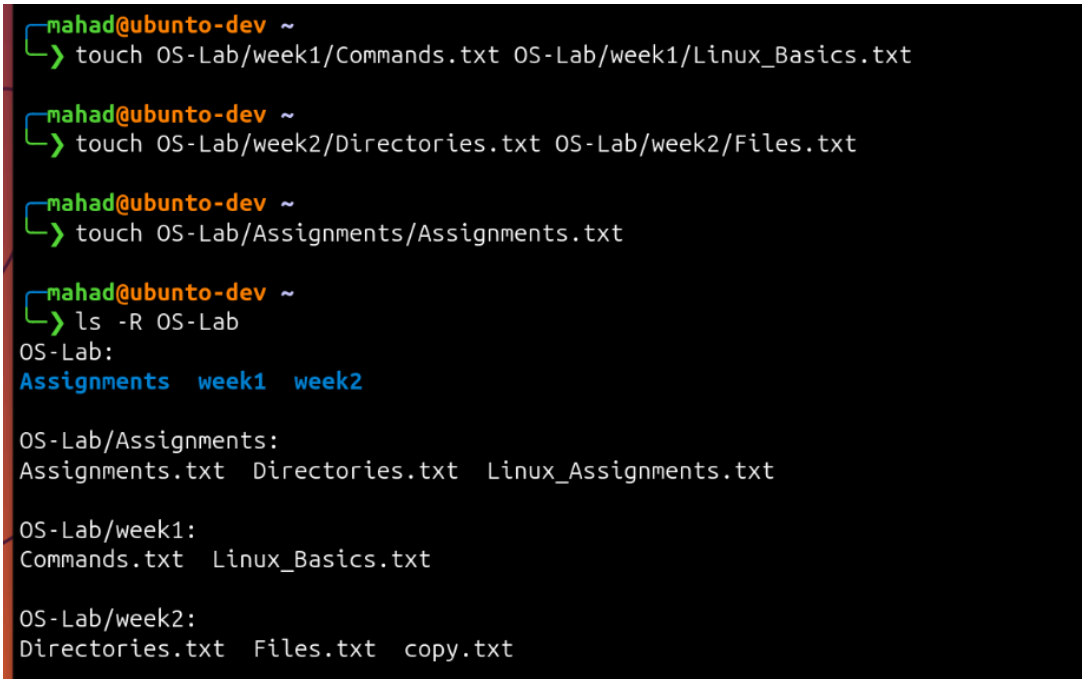
Create empty files inside the respective subdirectories: Commands.txt and Linux_Basics.txt in week1, Directories.txt and Files.txt in week2, and Assignments.txt in Assignments.

Commands:

```shell
touch OS-Lab/week1/Commands.txt OS-Lab/week1/Linux_Basics.txt
touch OS-Lab/week2/Directories.txt OS-Lab/week2/Files.txt
touch OS-Lab/Assignments/Assignments.txt
ls -R OS-Lab
```

The touch command creates empty target files in specified folder locations without changing directory context.

Output:



Task 4: Copying Files and Directories

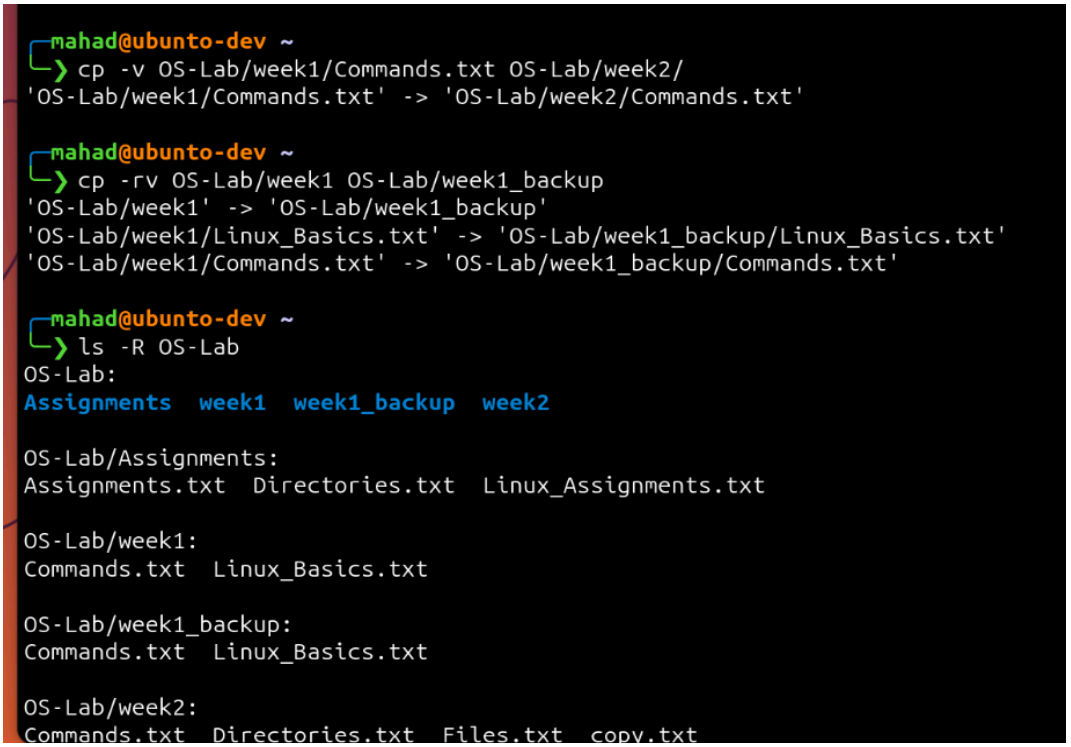
Copy Commands.txt from week1 to week2, and recursively copy the entire week1 directory to create a backup directory named week1_backup.

Commands:

```shell
cp -v OS-Lab/week1/Commands.txt OS-Lab/week2/
cp -rv OS-Lab/week1 OS-Lab/week1_backup
ls -R OS-Lab
```

The cp command copies individual files, while the -r flag enables recursive copying of entire directories and their contents.

Output:



Task 5: Moving and Renaming Files

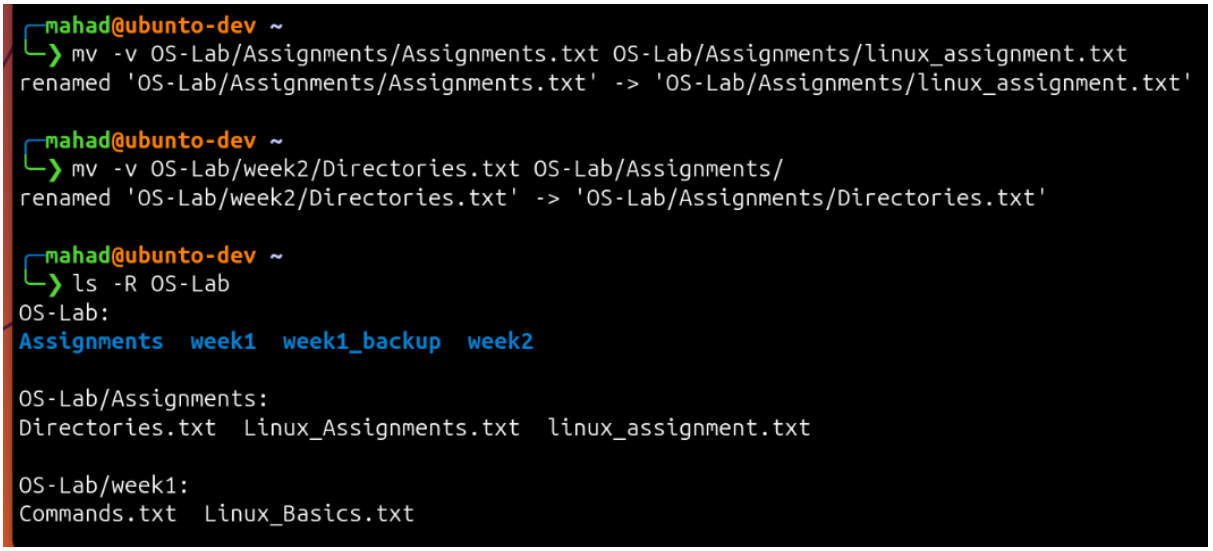
Rename Assignments.txt to linux_assignment.txt inside Assignments, and move Directories.txt from week2 into the Assignments directory.

Commands:

```shell
mv -v OS-Lab/Assignments/Assignments.txt
OS-Lab/Assignments/linux_assignment.txt
mv -v OS-Lab/week2/Directories.txt OS-Lab/Assignments/
ls -R OS-Lab
```

The mv command renames a file when destination is a new filename, and moves a file when destination is an existing directory.

Output:



Task 6: File Content Viewing and Editing

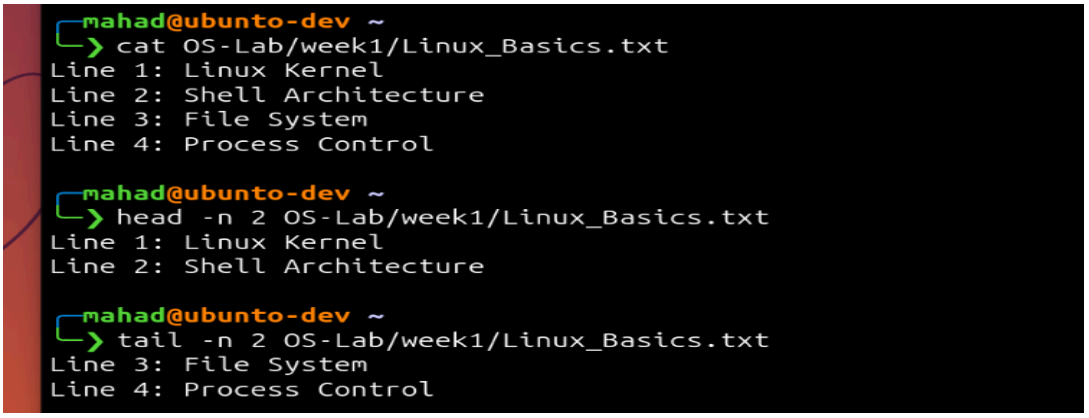
Populate Linux_Basics.txt with sample text, display the complete contents using cat, and view the first 2 and last 2 lines using head and tail.

Commands:

```
Shell
echo -e "Line 1: Linux Kernel\nLine 2: Shell Architecture\nLine 3: File
System\nLine 4: Process Control" > OS-Lab/week1/Linux_Basics.txt
cat OS-Lab/week1/Linux_Basics.txt
head -n 2 OS-Lab/week1/Linux_Basics.txt
tail -n 2 OS-Lab/week1/Linux_Basics.txt
```

The cat command displays full file content, while head -n and tail -n filter output to the specified starting or ending lines respectively.

Output:



Task 7: Deleting Files and Directories

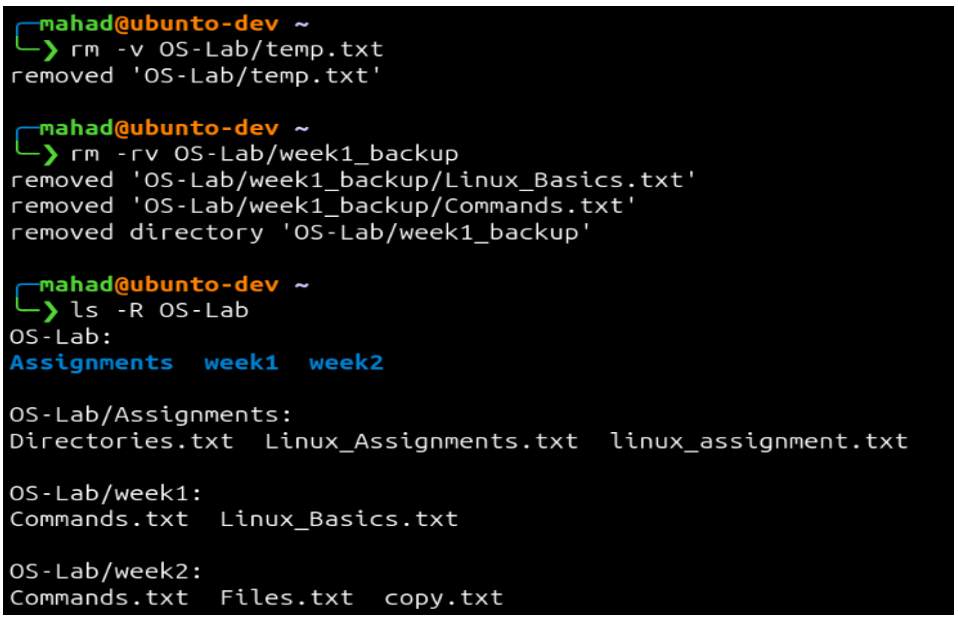
Create a temporary file temp.txt in OS-Lab, delete it, and then recursively remove the week1_backup directory.

Commands:

```shell
touch OS-Lab/temp.txt
rm -v OS-Lab/temp.txt
rm -rv OS-Lab/week1_backup
ls -R OS-Lab
```

The rm command removes individual files, whereas the -r option is required to recursively delete directories along with their nested items.

Output:



Task 8: File Searching and Finding

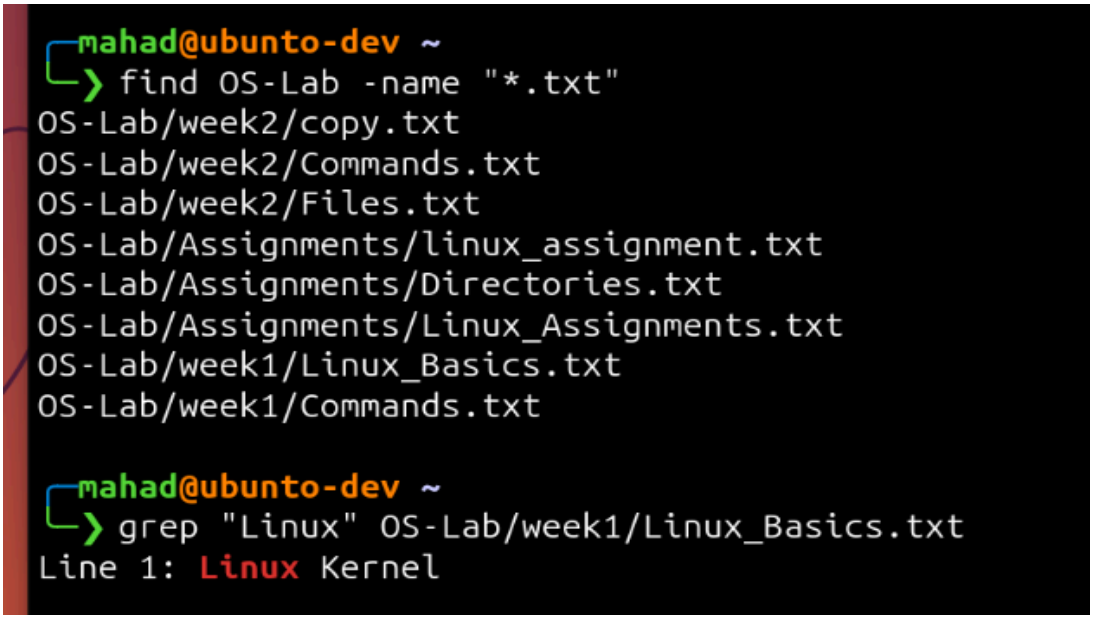
Search for all .txt files inside the OS-Lab directory structure, and locate occurrences of the string "Linux" within week1/Linux_Basics.txt.

Commands:

```shell
find OS-Lab -name "*.txt"
grep "Linux" OS-Lab/week1/Linux_Basics.txt
```

The find command searches for files matching a pattern within a directory tree, while grep searches for target text patterns inside file contents.

Output:



Task 9: Environment and Path Verification

Display the absolute path of the user's home directory, verify current user identity, and return to the home directory.

Commands:

```shell
whoami
echo $HOME
cd ~ && pwd
```

The whoami command outputs the current username, $HOME prints the environment variable for home directory path, and cd ~ ensures return to home folder.

Output:

